

STUDENT NUMBER: _____

North Sydney Boys High School
Year 12 Physics 2022
Trial HSC Examination

General Instructions

- Reading time – 5 minutes
- Working time – 135 minutes
- Board approved calculators may be used
- Write using black pen
- Write your student number at the top of every separated page
- Use the multiple-choice answer sheet provided at the end by separating it
- A NESA data and formula sheet is provided at the end and can be separated

Total Marks – 75

Part A - 16 marks

- Attempt Questions 1–16
- Allow about 20 minutes for this part

Part B - 59 marks

- Attempt Questions 17–24
- Allow about 115 minutes for this part

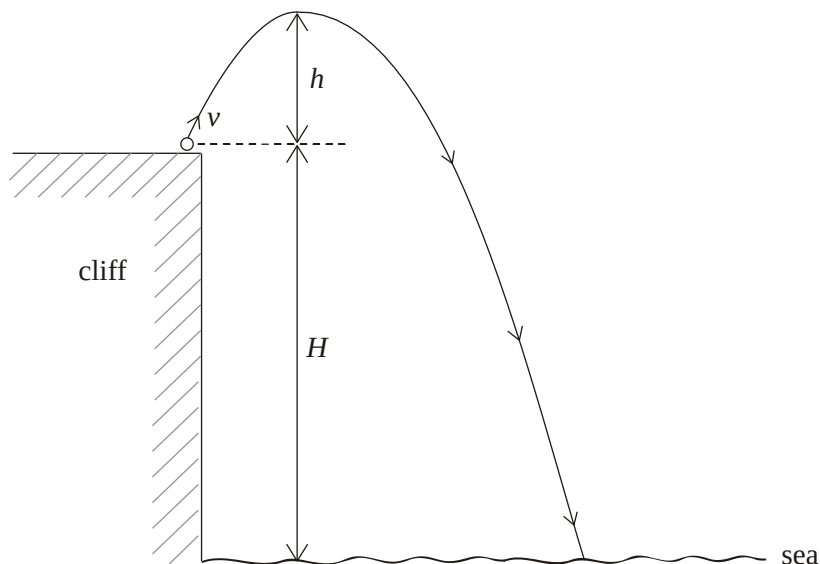
PART A /16	
PART B /59	
Total Mark /75	
KU /53	
WS /22	

This paper MUST NOT be removed from the examination room

Part A – 16 marks

Choose the most correct answer. Use the multiple-choice answer sheet for Questions 1–16

1. A stone is thrown with speed v from the top of a cliff of height H , as shown below.



The stone is thrown at an angle to the horizontal so that it rises to a height h above the top of the cliff before falling into the sea. The acceleration of free fall is g . Air resistance is negligible.

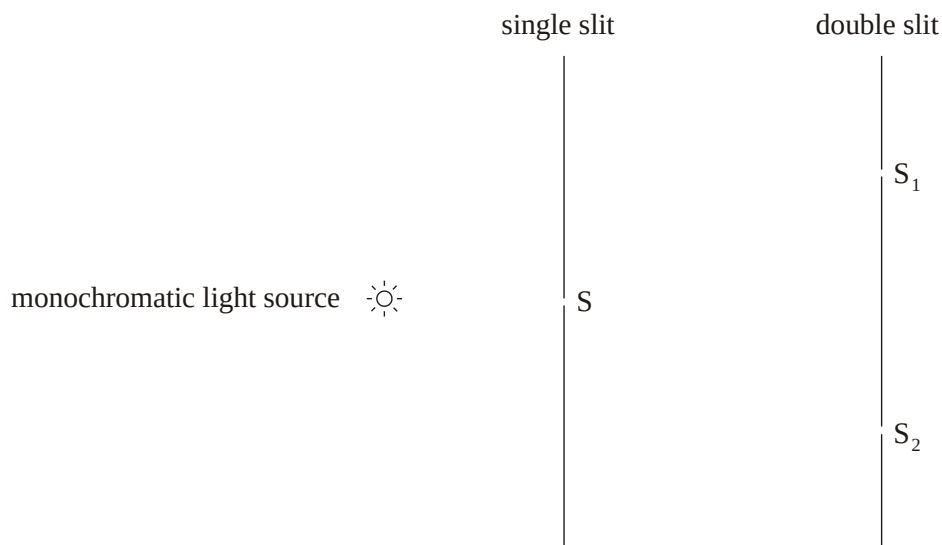
Which **one** of the following expressions correctly gives the speed of the stone as it hits the sea?

- A. $v + \sqrt{2gh}$
- B. $v + \sqrt{2gH}$
- C. $\sqrt{2g\{h+H\}}$
- D. $\sqrt{v^2 + 2gH}$
2. A transformer has a primary coil with N_p turns and a secondary coil with N_s turns. An alternating voltage supply of frequency f and voltage V_p is connected to the primary coil.

Which of the following correctly gives the frequency and voltage in the secondary coil?

	Frequency	Voltage
A.	$\frac{N_s}{N_p} f$	$\frac{N_p}{N_s} V_p$
B.	f	$\frac{N_p}{N_s} V_p$
C.	$\frac{N_p}{N_s} f$	$\frac{N_s}{N_p} V_p$
D.	f	$\frac{N_s}{N_p} V_p$

3. The rings of Saturn are made of rocky particles that orbit the planet. The period T of each particle depends on its distance r from the centre of Saturn. The period T is proportional to r^n . Which **one** of the following is n equal to?
- A. 1.0
- B. 1.5
- C. 2.0
- D. 3.0
4. The diagram below shows the arrangement for a Young's double slit experiment.



The function of the single slit is

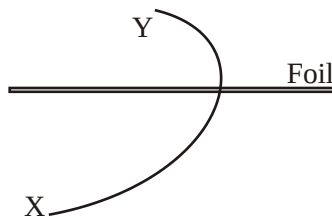
- A. to direct the light towards S_1 and S_2 .
- B. to ensure equal intensities of light at S_1 and S_2 .
- C. to produce coherent light at S_1 and S_2 .
- D. to reduce the intensity of light at S_1 and S_2 .
5. A proton of mass m and charge e is accelerated from rest through a potential difference V . The final speed of the proton is
- A. $\sqrt{\frac{2Ve}{m}}$.
- B. $\frac{2Ve}{m}$.
- C. $\sqrt{\frac{Ve}{m}}$.
- D. $\frac{Ve}{m}$.

6. Newton's law of gravitation for the force F between two point objects of masses M and m , separated by a distance d may be written as

$$Fd^2 \propto Mm.$$

The expression may also be used for the force of attraction between the Sun and the Earth, although they are not point masses. This is because

- A. the gravitational constant G is not involved in the expression.
 B. the force between the Sun and the Earth is very large.
 C. the separation of the Sun and the Earth is much greater than their radii.
 D. the mass of the Earth is much less than the mass of the Sun.
7. The diagram below shows the path XY of an electron that passes through a thin foil placed in a vacuum.



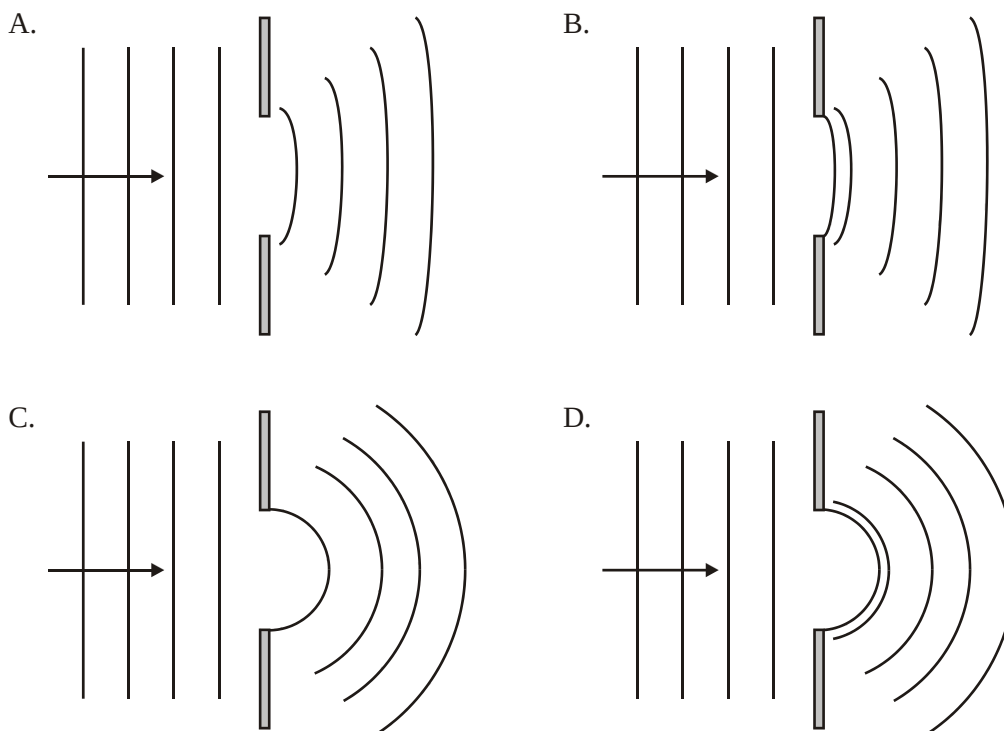
There is a uniform magnetic field acting perpendicularly to the plane of the page. Which **one** of the following statements best explains the path followed by the electron?

	Direction of motion of the electron	Direction of the magnetic field
A.	Y to X	into the plane of the page
B.	Y to X	out of the plane of the page
C.	X to Y	into the plane of the page
D.	X to Y	out of the plane of the page

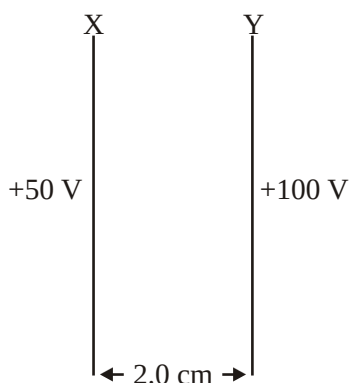
8. A satellite is in orbit about Earth. The satellite moves to an orbit closer to Earth. Which of the following correctly gives the change in the potential energy and the kinetic energy of the satellite?

	change in potential energy	change in kinetic energy
A.	Decreases	Increases
B.	Decreases	Decreases
C.	Increases	Increases
D.	Increases	Decreases

9. Which diagram best shows diffraction of plane wavefronts at a single slit?



10. The diagram below shows two lines of equipotential in a region of a uniform electric field. Line X has a potential of +50 V and line Y has a potential of +100V. The distance between X and Y is 2.0 cm.



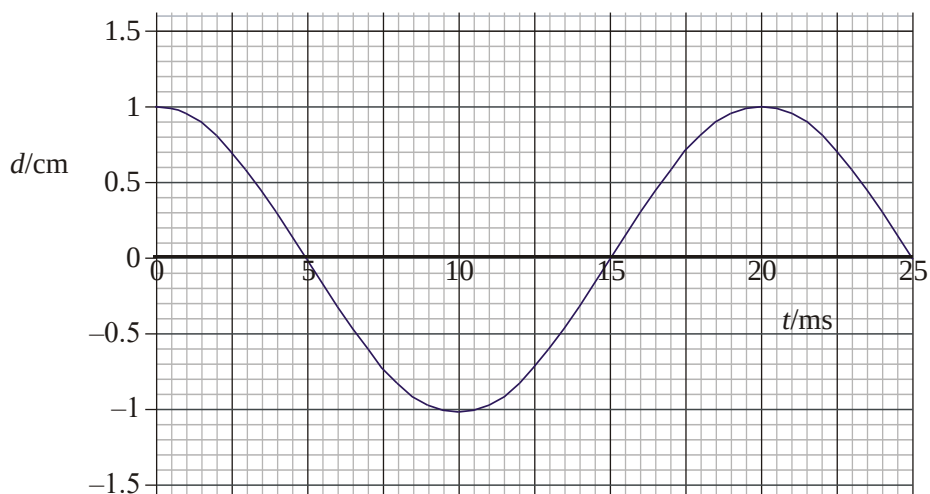
Which **one** of the following correctly gives the direction of the electric field and its strength?

	Direction	Strength / V cm^{-1}
A.	$X \rightarrow Y$	25
B.	$X \rightarrow Y$	100
C.	$Y \rightarrow X$	25
D.	$Y \rightarrow X$	100

11. Two satellites of equal mass, S_1 and S_2 , orbit the Earth. S_1 is orbiting at a distance r from the Earth's centre at speed v . S_2 orbits at a distance $2r$ from the Earth's centre at speed $\frac{v}{\sqrt{2}}$. The ratio of the centripetal force on S_1 to the centripetal force on S_2 is

- A. $\frac{1}{8}$.
 B. $\frac{1}{4}$.
 C. 4.
 D. 8.

12. A wave is travelling through a medium. The diagram shows the variation with time t of the displacement d of a particle of the medium from $t = 0$ to $t = 25$ ms.



Which of the following correctly gives the frequency and the amplitude of the wave?

	frequency / Hz	amplitude / cm
A.	2.0×10^{-2}	2.0
B.	2.0×10^{-2}	1.0
C.	50	2.0
D.	50	1.0

13. Which **one** of the following is a correct definition of electric potential difference between two points?
- A. The power to move a small positive charge between the two points.
 - B. The work done to move a small positive charge between the two points.
 - C. The power per unit charge to move a small positive charge between the two points.
 - D. The work done per unit charge to move a small positive charge between the two points.
14. The acceleration of free fall of a small sphere of mass $5.0 \times 10^{-3} \text{ kg}$ when close to the surface of Jupiter is 25 ms^{-2} . The gravitational field strength at the surface of Jupiter is
- A. $2.0 \times 10^{-4} \text{ N kg}^{-1}$.
 - B. $1.3 \times 10^{-1} \text{ N kg}^{-1}$.
 - C. 25 N kg^{-1} .
 - D. $5.0 \times 10^3 \text{ N kg}^{-1}$.
15. The waves from two light sources meet at a point. Which condition is essential for interference to be observed?
- A. Constant phase difference between the waves
 - B. Equal amplitude of the waves
 - C. Equal frequency of the waves
 - D. Equal intensities of the waves
16. The escape speed of an object of mass m from a planet of mass M and radius r depends on the gravitational constant and
- A. M and r .
 - B. m and r .
 - C. M only.
 - D. M , m , and r .

Part B – 59 marks

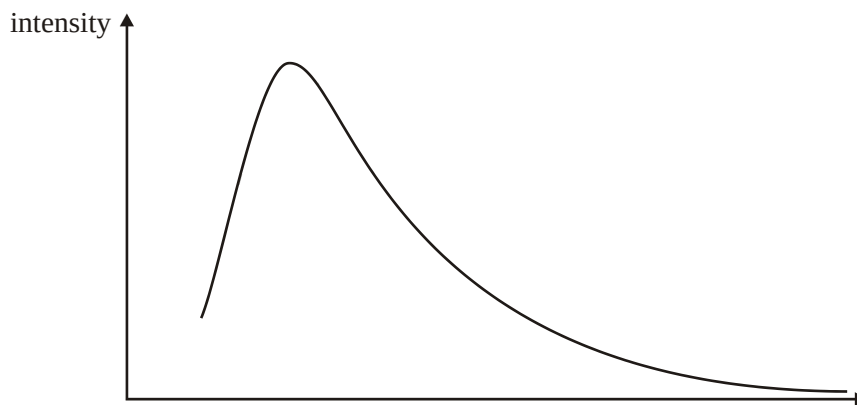
Write your answers in the spaces provided

17. (a) Explain the term *black-body radiation*.

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(1)

The diagram below is a sketch graph of the black-body radiation spectrum of a certain star.



- (b) Label the x-axis of the graph.

(1)

- (c) On the graph, sketch the black-body radiation spectrum of a star that has a lower surface temperature and lower apparent brightness than this star.

(1)

The star Betelgeuse in the Orion constellation emits black-body radiation that has a maximum intensity at a wavelength of $0.97\ \mu\text{m}$.

- (d) Deduce that the surface temperature of Betelgeuse is about 3000 K.

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(1)

(Total 4 marks)

18. (a) State what is meant by polarised light.

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(1)

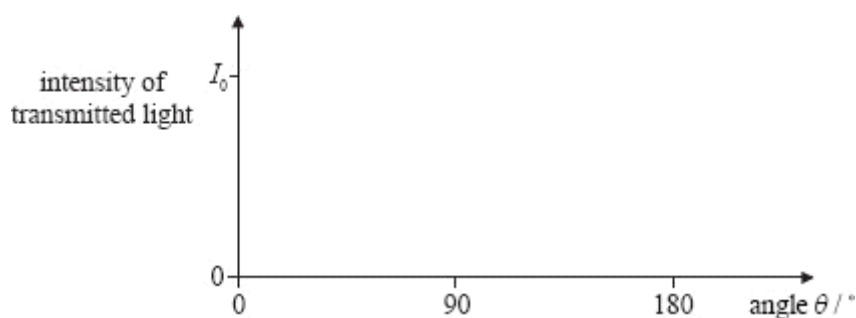
- (b) Polarised light of intensity I_0 is incident on an analyser. The transmission axis of the analyser makes an angle θ with the direction of the electric field of the light.

- (i) Calculate, in terms of I_0 , the intensity of light transmitted through the analyser when $\theta = 60^\circ$.

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(1)

- (ii) On the axes below, sketch a graph to show the variation with angle θ of the intensity of the transmitted light.



(2)

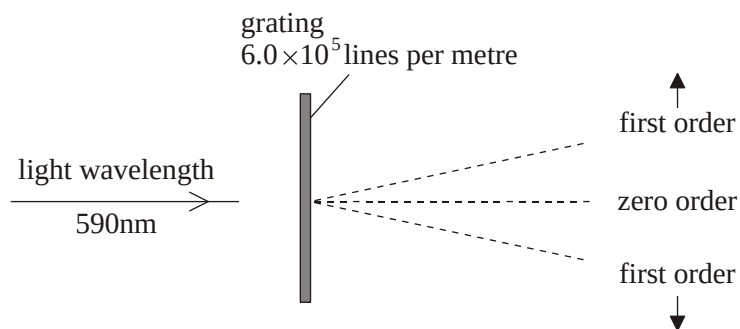
- (c) Outline how polarising sunglasses reduce glare from a reflecting surface.

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(2)

(Total 6 marks)

19. Light of wavelength 590 nm is incident normally on a diffraction grating, as shown below.



The grating has 6.0×10^5 lines per metre.

- (a) Determine the **total** number of orders (both sides) of diffracted light, including the zero order, that can be observed.

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(4)

- (b) The incident light is replaced by a beam of light consisting of two wavelengths, 590 nm and 589 nm.

State **two** observable differences between a first order spectrum and a second order spectrum of the diffracted light.

1.
-
2.
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(2)

(Total 6 marks)

20. (a) A satellite goes round a planet at constant speed. Explain why, although its speed is constant, it is accelerating.

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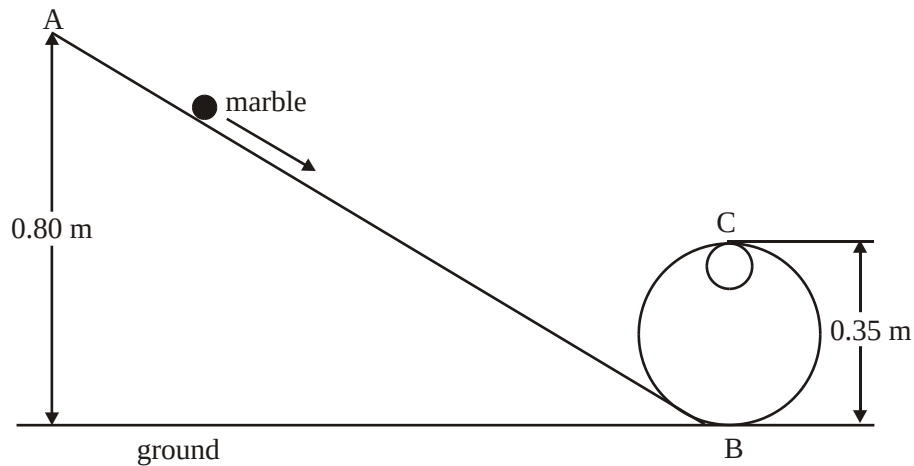
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(1)

In the diagram below, a marble (small glass sphere) rolls down a track, the bottom part of which has been bent into a loop. The end A of the track, from which the marble is released, is at a height of 0.80 m above the ground. Point B is the lowest point and point C the highest point of the loop. The diameter of the loop is 0.35 m.



The mass of the marble is 0.050 kg. Friction forces and any gain in kinetic energy due to the rotating of the marble can be ignored.

Consider the marble when it is at point C.

- (b) (i) On the diagram above, draw an arrow to show the direction of the resultant force acting on the marble.

(1)

- (ii) State the names of the **two** forces acting on the marble.

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(2)

- (iii) Deduce that the speed of the marble is approximately 3.0 ms^{-1} .

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(2)

- (iv) Determine the resultant force acting on the marble and hence determine the reaction force of the track on the marble.

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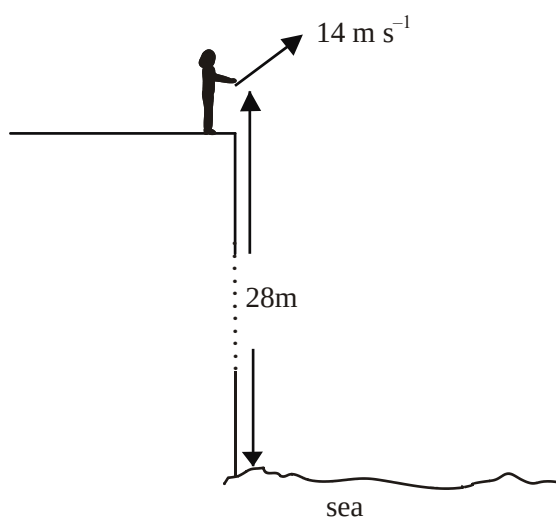
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(3)

(Total 9 marks)

21. A stone is thrown from the top of a cliff of height 28 m above the sea. The stone is thrown at a speed of 14 m s^{-1} at an angle above the horizontal. Air resistance is negligible.



The maximum height reached by the stone measured from the point from which it is thrown is 8.0 m.

- (a) By considering the energy of the stone, determine the speed with which the stone hits the sea.

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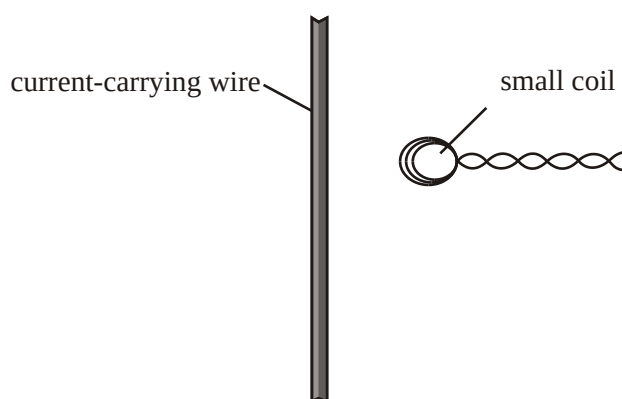
(2)

- (b) The stone leaves the cliff at time $t = 0$. It reaches its maximum height at $t = T_H$. On the axis below, draw a sketch-graph to show the variation with time t of the magnitude of the **vertical** component of speed of the stone from $t = 0$ to $t = T_S$, the time just before the stone strikes the sea.



(2)
(Total 4 marks)

22. A small coil is placed with its plane parallel to a long straight current-carrying wire, as shown below.



- (a) (i) State Faraday's law of electromagnetic induction.

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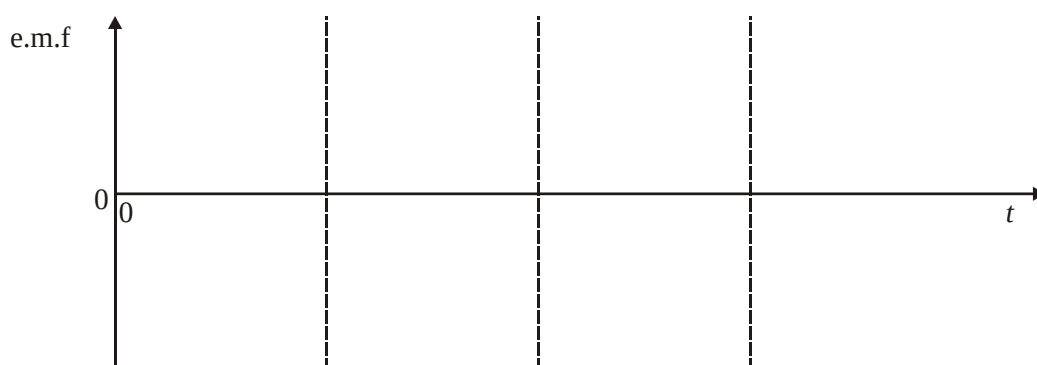
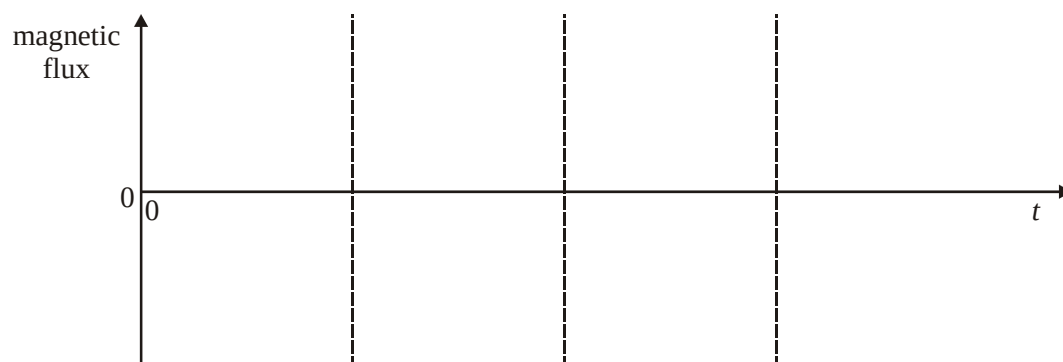
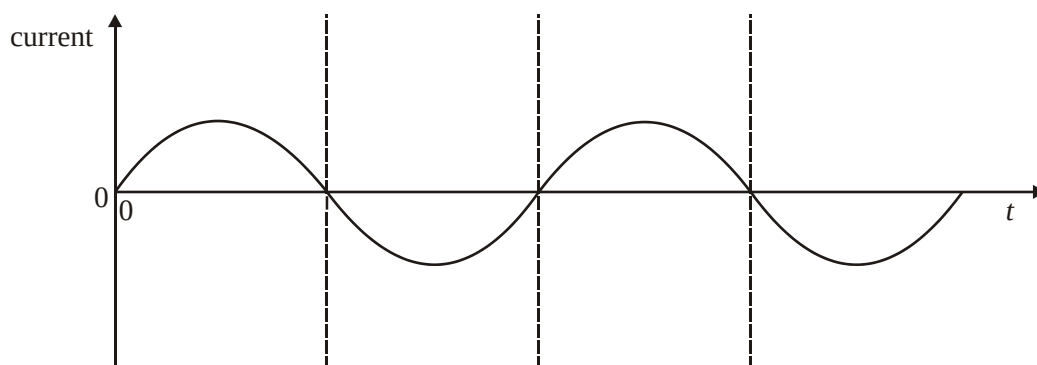
(2)

- (ii) Use the law to explain why, when the current in the wire changes, an emf is induced in the coil.

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(1)

The diagram below shows the variation with time t of the current in the wire.



- (b) (i) Draw, on the axes provided, a sketch-graph to show the variation with time t of the magnetic flux in the coil. (1)

- (ii) Construct, on the axes provided, a sketch-graph to show the variation with time t of the emf induced in the coil. (2)

- (iii) State and explain the effect on the maximum emf induced in the coil when the coil is further away from the wire.

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(2)

- (c) Such a coil may be used to measure large alternating currents in a high-voltage cable. Identify **one** advantage and **one** disadvantage of this method.

Advantage:

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Disadvantage:

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(2)
(Total 10 marks)

23. This question is about Kepler's third law.

- (a) Kepler's third law states that the period T of the orbit of a planet about the Sun is related to the average orbital radius R of the planet by the relationship

$$T^2 = KR^3$$

where K is a constant.

- (i) Suggest why the law specifies the average orbital radius.

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(1)

- (ii) State the name of the force that causes the acceleration of the planets orbiting the Sun.

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(1)

- (iii) State an expression for the magnitude F of the force in (ii) in terms of the mass M_S , of the Sun, the mass m of the planet, the radius R of the orbit and the universal gravitational constant G .

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(1)

- (iv) Hence deduce, explaining your working, that the constant K is given by the expression

$$K = \frac{4\pi^2}{GM_S}$$

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(4)

- (b) Ganymede is one of the moons of Jupiter and the following data are available.

Average orbital radius of Ganymede = 1.1×10^9 m

Orbital period of Ganymede = 6.2×10^5 s

Universal gravitational constant $G = 6.67 \times 10^{-11}$ N m² kg⁻²

- (i) Deduce that the gravitational field strength of Jupiter at the surface of Ganymede is approximately 0.1 N kg^{-1} .

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(2)

- (ii) Estimate the mass of Jupiter.

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(3)

(Total 12 marks)

24. (a) Describe one historical method used to determine the speed of light experimentally including its limitations.

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[illegible]

- (b) In 1983, the speed of light was declared to be exactly $299\,792\,458\text{ ms}^{-1}$. Explain the currently accepted relationship of the speed of light to the measurement of distance and time.

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(2)
(Total 8 marks)

End of paper

DATA SHEET

FORMULAE SHEET

		Motion, forces and gravity
Charge on electron, q_e	$-1.602 \times 10^{-19} \text{ C}$	$s = ut + \frac{1}{2}at^2$
Mass of electron, m_e	$9.109 \times 10^{-31} \text{ kg}$	$\vec{F}_{\text{net}} = m\vec{a}$
Mass of neutron, m_n	$1.675 \times 10^{-27} \text{ kg}$	$W = F_{\parallel}s = F_s \cos \theta$
Mass of proton, m_p	$1.673 \times 10^{-27} \text{ kg}$	$K = \frac{1}{2}mv^2$
Speed of sound in air	340 m s^{-1}	$P = F_{\parallel}v = F_v \cos \theta$
Earth's gravitational acceleration, g	9.8 m s^{-2}	$\sum \frac{1}{2}mv^2_{\text{before}} = \sum \frac{1}{2}mv^2_{\text{after}}$
Speed of light, c	$3.00 \times 10^8 \text{ m s}^{-1}$	$a_c = \frac{v^2}{r}$
Electric permittivity constant, ϵ_0	$8.854 \times 10^{-12} \text{ A}^2 \text{ s}^4 \text{ kg}^{-1} \text{ m}^{-3}$	$F_c = \frac{mv^2}{r}$
Magnetic permeability constant, μ_0	$4\pi \times 10^{-7} \text{ N A}^{-2}$	$F = \frac{GMm}{r^2}$
Universal gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$	$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}$
Mass of Earth, M_E	$6.0 \times 10^{24} \text{ kg}$	
Radius of Earth, r_E	$6.371 \times 10^6 \text{ m}$	
Planck constant, h	$6.626 \times 10^{-34} \text{ J s}$	
Rydberg constant, R (hydrogen)	$1.097 \times 10^7 \text{ m}^{-1}$	$f_{\text{beat}} = f_2 - f_1 $
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg}$	$f' = f \frac{(v_{\text{wave}} + v_{\text{observer}})}{(v_{\text{wave}} - v_{\text{source}})}$
1 eV	$931.5 \text{ MeV}/c^2$	$n_1 \sin \theta_1 = n_2 \sin \theta_2$
Density of water, ρ	$1.602 \times 10^{-19} \text{ J}$	$\sin \theta_c = \frac{n_2}{n_1}$
Specific heat capacity of water	$1.00 \times 10^3 \text{ kg m}^{-3}$	$I_1 r_1^2 = I_2 r_2^2$
Wien's displacement constant, b	$4.18 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$	$\frac{Q}{t} = \frac{kA\Delta T}{d}$
	$2.898 \times 10^{-3} \text{ m K}$	
		Waves and thermodynamics
		$v = f\lambda$
		$f = \frac{1}{T}$
		$d \sin \theta = m\lambda$
		$n_x = \frac{c}{v_x}$
		$I = I_{\text{max}} \cos^2 \theta$
		$Q = mc\Delta T$

Electricity and magnetism

$$E = \frac{V}{d}$$

$$V = \frac{\Delta U}{q}$$

$$W = qV$$

$$W = qEd$$

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B = \frac{\mu_0 NI}{L}$$

$$\Phi = B_{\parallel}A = BA\cos\theta$$

$$\varepsilon = -N\frac{\Delta\Phi}{\Delta t}$$

$$\frac{V_p}{V_s} = \frac{N_p}{N_s}$$

Quantum, special relativity and nuclear

$$\lambda = \frac{h}{mv}$$

$$K_{\max} = hf - \phi$$

$$\lambda_{\max} = \frac{b}{T}$$

$$E = mc^2$$

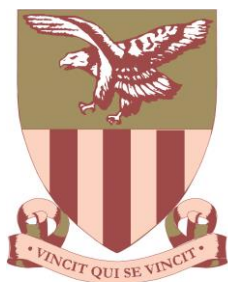
$$E = hf$$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

PERIODIC TABLE OF THE ELEMENTS

1 H 1.008	2 He 4.003	KEY																Atomic Number Symbol Weight Name Gold																Periodic Table of the Elements																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
3 Li 6.941	4 Be 9.012	Boron Carbon Nitrogen Oxygen Fluorine Neon																Sodium Magnesium Aluminum Silicon Phosphorus Sulfur Chlorine Argon																Potassium Calcium Scandium Titanium Vanadium Chromium Manganese Iron Cobalt Nickel Copper Zinc Gallium Germanium Arsenic Selenium Bromine Krypton																Francium Radium Actinoids Lanthanoids Cerium Praseodymium Neodymium Promethium Samarium Europium Gadolinium Terbium Dysprosium Holmium Erbium Thulium Ytterbium Lutetium																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18	11 Na 22.99	12 Mg 24.31	13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.07	17 Cl 35.45	18 Ar 39.95	19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.87	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.38	31 Ga 69.72	32 Ge 72.64	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80	37 Rb 85.47	38 Sr 87.61	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.96	43 Tc 101.1	44 Ru 102.9	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3	55 Cs 132.9	56 Ba 137.3	57 La 138.9	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm 144.9	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.1	71 Lu 175.0	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po 209.0	85 At 210.0	86 Rn 222.0	87 Fr 223.0	88 Ra 226.0	89-103 Actinoids	104 Rf 261.1	105 Db 262.1	106 Sg 266.1	107 Bh 264.1	108 Hs 277.1	109 Mt 268.1	110 Ds 271.1	111 Rg 272.1	112 Cn 285.1	113 Nh 284.1	114 Fl 289.1	115 Mc 288.1	116 Lv 293.1	117 Ts 294.1	118 Og 294.1	119 Uu 295.1	120 Uu 296.1	121 Uu 297.1	122 Uu 298.1	123 Uu 299.1	124 Uu 300.1	125 Uu 301.1	126 Uu 302.1	127 Uu 303.1	128 Uu 304.1	129 Uu 305.1	130 Uu 306.1	131 Uu 307.1	132 Uu 308.1	133 Uu 309.1	134 Uu 310.1	135 Uu 311.1	136 Uu 312.1	137 Uu 313.1	138 Uu 314.1	139 Uu 315.1	140 Uu 316.1	141 Uu 317.1	142 Uu 318.1	143 Uu 319.1	144 Uu 320.1	145 Uu 321.1	146 Uu 322.1	147 Uu 323.1	148 Uu 324.1	149 Uu 325.1	150 Uu 326.1	151 Uu 327.1	152 Uu 328.1	153 Uu 329.1	154 Uu 330.1	155 Uu 331.1	156 Uu 332.1	157 Uu 333.1	158 Uu 334.1	159 Uu 335.1	160 Uu 336.1	161 Uu 337.1	162 Uu 338.1	163 Uu 339.1	164 Uu 340.1	165 Uu 341.1	166 Uu 342.1	167 Uu 343.1	168 Uu 344.1	169 Uu 345.1	170 Uu 346.1	171 Uu 347.1	172 Uu 348.1	173 Uu 349.1	174 Uu 350.1	175 Uu 351.1	176 Uu 352.1	177 Uu 353.1	178 Uu 354.1	179 Uu 355.1	180 Uu 356.1	181 Uu 357.1	182 Uu 358.1	183 Uu 359.1	184 Uu 360.1	185 Uu 361.1	186 Uu 362.1	187 Uu 363.1	188 Uu 364.1	189 Uu 365.1	190 Uu 366.1	191 Uu 367.1	192 Uu 368.1	193 Uu 369.1	194 Uu 370.1	195 Uu 371.1	196 Uu 372.1	197 Uu 373.1	198 Uu 374.1	199 Uu 375.1	200 Uu 376.1	201 Uu 377.1	202 Uu 378.1	203 Uu 379.1	204 Uu 380.1	205 Uu 381.1	206 Uu 382.1	207 Uu 383.1	208 Uu 384.1	209 Uu 385.1	210 Uu 386.1	211 Uu 387.1	212 Uu 388.1	213 Uu 389.1	214 Uu 390.1	215 Uu 391.1	216 Uu 392.1	217 Uu 393.1	218 Uu 394.1	219 Uu 395.1	220 Uu 396.1	221 Uu 397.1	222 Uu 398.1	223 Uu 399.1	224 Uu 400.1	225 Uu 401.1	226 Uu 402.1	227 Uu 403.1	228 Uu 404.1	229 Uu 405.1	230 Uu 406.1	231 Uu 407.1	232 Uu 408.1	233 Uu 409.1	234 Uu 410.1	235 Uu 411.1	236 Uu 412.1	237 Uu 413.1	238 Uu 414.1	239 Uu 415.1	240 Uu 416.1	241 Uu 417.1	242 Uu 418.1	243 Uu 419.1	244 Uu 420.1	245 Uu 421.1	246 Uu 422.1	247 Uu 423.1	248 Uu 424.1	249 Uu 425.1	250 Uu 426.1	251 Uu 427.1	252 Uu 428.1	253 Uu 429.1	254 Uu 430.1	255 Uu 431.1	256 Uu 432.1	257 Uu 433.1	258 Uu 434.1	259 Uu 435.1	260 Uu 436.1	261 Uu 437.1	262 Uu 438.1	263 Uu 439.1	264 Uu 440.1	265 Uu 441.1	266 Uu 442.1	267 Uu 443.1	268 Uu 444.1	269 Uu 445.1	270 Uu 446.1	271 Uu 447.1	272 Uu 448.1	273 Uu 449.1	274 Uu 450.1	275 Uu 451.1	276 Uu 452.1	277 Uu 453.1	278 Uu 454.1	279 Uu 455.1	280 Uu 456.1	281 Uu 457.1	282 Uu 458.1	283 Uu 459.1	284 Uu 460.1	285 Uu 461.1	286 Uu 462.1	287 Uu 463.1	288 Uu 464.1	289 Uu 465.1	290 Uu 466.1	291 Uu 467.1	292 Uu 468.1	293 Uu 469.1	294 Uu 470.1	295 Uu 471.1	296 Uu 472.1	297 Uu 473.1	298 Uu 474.1	299 Uu 475.1	300 Uu 476.1	301 Uu 477.1	302 Uu 478.1	303 Uu 479.1	304 Uu 480.1	305 Uu 481.1	306 Uu 482.1	307 Uu 483.1	308 Uu 484.1	309 Uu 485.1	310 Uu 486.1	311 Uu 487.1	312 Uu 488.1	313 Uu 489.1	314 Uu 490.1	315 Uu 491.1	316 Uu 492.1	317 Uu 493.1	318 Uu 494.1	319 Uu 495.1	320 Uu 496.1	321 Uu 497.1	322 Uu 498.1	323 Uu 499.1	324 Uu 500.1	325 Uu 501.1	326 Uu 502.1	327 Uu 503.1	328 Uu 504.1	329 Uu 505.1	330 Uu 506.1	331 Uu 507.1	332 Uu 508.1	333 Uu 509.1	334 Uu 510.1	335 Uu 511.1	336 Uu 512.1	337 Uu 513.1	338 Uu 514.1	339 Uu 515.1	340 Uu 516.1	341 Uu 517.1	342 Uu 518.1	343 Uu 519.1	344 Uu 520.1	345 Uu 521.1	346 Uu 522.1	347 Uu 523.1	348 Uu 524.1	349 Uu 525.1	350 Uu 526.1	351 Uu 527.1	352 Uu 528.1	353 Uu 529.1	354 Uu 530.1	355 Uu 531.1	356 Uu 532.1	357 Uu 533.1	358 Uu 534.1	359 Uu 535.1	360 Uu 536.1	361 Uu 537.1	362 Uu 538.1	363 Uu 539.1	364 Uu 540.1	365 Uu 541.1	366 Uu 542.1	367 Uu 543.1	368 Uu 544.1	369 Uu 545.1	370 Uu 546.1	371 Uu 547.1	372 Uu 548.1	373 Uu 549.1	374 Uu 550.1	375 Uu 551.1	376 Uu 552.1	377 Uu 553.1	378 Uu 554.1	379 Uu 555.1	380 Uu 556.1	381 Uu 557.1	382 Uu 558.1	383 Uu 559.1	384 Uu 560.1	385 Uu 561.1	386 Uu 562.1	387 Uu 563.1	388 Uu 564.1	389 Uu 565.1	390 Uu 566.1	391 Uu 567.1	392 Uu 568.1	393 Uu 569.1	394 Uu 570.1	395 Uu 571.1	396 Uu 572.1	397 Uu 573.1	398 Uu 574.1	399 Uu 575.1	400 Uu 576.1	401 Uu 577.1	402 Uu 578.1	403 Uu 579.1	404 Uu 580.1	405 Uu 581.1	406 Uu 582.1	407 Uu 583.1	408 Uu 584.1	409 Uu 585.1	410 Uu 586.1	411 Uu 587.1	412 Uu 588.1	413 Uu 589.1	414 Uu 590.1	415 Uu 591.1	416 Uu 592.1	417 Uu 593.1	418 Uu 594.1	419 Uu 595.1	420 Uu 596.1	421 Uu 597.1	422 Uu 598.1	423 Uu 599.1	424 Uu 600.1	425 Uu 601.1	426 Uu 602.1	427 Uu 603.1	428 Uu 604.1	429 Uu 605.1	430 Uu 606.1	431 Uu 607.1	432 Uu 608.1	433 Uu 609.1	434 Uu 610.1	435 Uu 611.1	436 Uu 612.1	437 Uu 613.1	438 Uu 614.1	439 Uu 615.1	440 Uu 616.1	441 Uu 617.1	442 Uu 618.1	443 Uu 619.1	444 Uu 620.1	445 Uu 621.1	446 Uu 622.1	447 Uu 623.1	448 Uu 624.1	449 Uu 625.1	450 Uu 626.1	451 Uu 627.1	452 Uu 628.1	453 Uu 629.1	454 Uu 630.1	455 Uu 631.1	456 Uu 632.1	457 Uu 633.1	458 Uu 634.1	459 Uu 635.1	460 Uu 636.1	461 Uu 637.1	462 Uu 638.1	463 Uu 639.1	464 Uu 640.1	465 Uu 641.1	466 Uu 642.1	467 Uu 643.1	468 Uu 644.1	469 Uu 645.1	470 Uu 646.1	471 Uu 647.1	472 Uu 648.1	473 Uu 649.1	474 Uu 650.1	475 Uu 651.1	476 Uu 652.1	477 Uu 653.1	478 Uu 654.1	479 Uu 655.1	480 Uu 656.1	481 Uu 657.1	482 Uu 658.1	483 Uu 659.1	484 Uu 660.1	485 Uu 661.1	486 Uu 662.1	487 Uu 663.1	488 Uu 664.1	489 Uu 665.1	490 Uu 666.1	491 Uu 667.1	492 Uu 668.1	493 Uu 669.1	494 Uu 670.1	495 Uu 671.1	496 Uu 672.1	497 Uu 673.1	498 Uu 674.1	499 Uu 675.1	500 Uu 676.1	501 Uu 677.1	502 Uu 678.1	503 Uu 679.1	504 Uu 680.1	505 Uu 681.1	506 Uu 682.1	507 Uu 683.1	508 Uu 684.1	509 Uu 685.1	510 Uu 686.1	511 Uu 687.1	512 Uu 688.1	513 Uu 689.1	514 Uu 690.1	515 Uu 691.1	516 Uu 692.1	517 Uu 693.1	518 Uu 694.1	519 Uu 695.1	520 Uu 696.1	521 Uu 697.1	522 Uu 698.1	523 Uu 699.1	524 Uu 700.1	525 Uu 701.1	526 Uu 702.1	527 Uu 703.1	528 Uu 704.1	529 Uu 705.1	530 Uu 706.1	531 Uu 707.1	532 Uu 708.1	533 Uu 709.1	534 Uu 710.1	535 Uu 711.1	536 Uu 712.1	537 Uu 713.1	538 Uu 714.1	539 Uu 715.1	540 Uu 716.1	541 Uu 717.1	542 Uu 718.1	543 Uu 719.1	544 Uu 720.1	545 Uu 721.1	546 Uu 722.1	547 Uu 723.1	548 Uu 724.1	549 Uu 725.1	550 Uu 726.1	551 Uu 727.1	552 Uu 728.1	553 Uu 729.1	554 Uu 730.1	555 Uu 731.1	556 Uu 732.1	557 Uu 733.1	558 Uu 734.1	559 Uu 735.1	560 Uu 736.1	561 Uu 737.1	562 Uu 738.1	563 Uu 739.1	564 Uu 740.1	565 Uu 741.1	566 Uu 742.1	567 Uu 743.1	568 Uu 744.1	569 Uu 745.1	570 Uu 746.1	571 Uu 747.1	572 Uu 748.1	573 Uu 749.1	574 Uu 750.1	575 Uu 751.1	576 Uu 752.1	577 Uu 753.1	578 Uu 754.1	579 Uu 755.1	580 Uu 756.1	581 Uu 757.1	582 Uu 758.1	583 Uu 759.1	584 Uu 760.1	585 Uu 761.1	586 Uu 762.1	587 Uu 763.1	588 Uu 764.1	589 Uu 765.1	590 Uu 766.1	591 Uu 767.1	592 Uu 768.1	593 Uu 769.1	594 Uu 770.1	595 Uu 771.1	596 Uu 772.1	597 Uu 773.1	598 Uu 774.1	599 Uu 775.1	600 Uu 776.1	601 Uu 777.1	602 Uu 778.1	603 Uu 779.1	604 Uu 780.1	605 Uu 781.1	606 Uu 782.1	607 Uu 783.1	608 Uu 784.1	609 Uu 785.1	610 Uu 786.1	611 Uu 787.1	612 Uu 788.1	613 Uu 789.1	614 Uu 790.1	615 Uu 791.1	616 Uu 792.1	617 Uu 793.1	618 Uu 794.1	619 Uu 795.1	620 Uu 796.1	621 Uu 797.1	622 Uu 798.1	623 Uu 799.1	624 Uu 800.1	625 Uu 801.1	626 Uu 802.1	627 Uu 803.1	628 Uu 804.1	629 Uu 805.1	630 Uu 806.1	631 Uu 807.1	632 Uu 808.1	633 Uu 809.1	634 Uu 810.1	635 Uu 811.1	636 Uu 812.1	637 Uu 813.1	638 Uu 814.1	639 Uu 815.1	640 Uu 816.1	641 Uu 817.1	642 Uu 818.1	643 Uu 819.1	644 Uu 820.1	645 Uu 821.1	646 Uu 822.1	647 Uu 823.1	648 Uu 824.1	649 Uu 825.1	650 Uu 826.1	651 Uu 827.1	652 Uu 828.1	653 Uu 829.1	654 Uu 830.1	655 Uu 831.1	656 U

STUDENT NUMBER: _____



North Sydney Boys High School

Science Faculty

HSC Physics 2022

Assessment 3

MULTIPLE CHOICE ANSWER SHEET

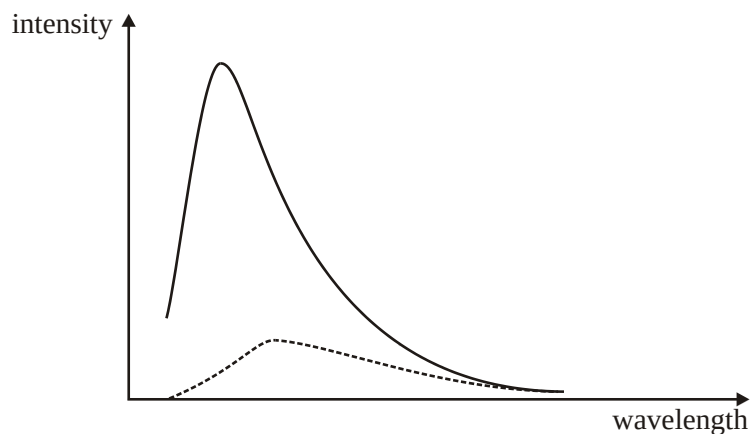
- | | | | | | | | | |
|-----|---|-----------------------|---|-----------------------|---|-----------------------|---|-----------------------|
| 1. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 2. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 3. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 4. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 5. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 6. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 7. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 8. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 9. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 10. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 11. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 12. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 13. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 14. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 15. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |
| 16. | A | ☐ | B | ☐ | C | ☐ | D | ☐ |

Marking Guideline Year 12 Physics Trial Exam 2022

Note: Working Scientifically marks have (WS)

- | | | |
|-----|---|---|
| 1. | D | 1 |
| 2. | D | |
| 3. | B | |
| 4. | C | |
| 5. | A | |
| 6. | C | |
| 7. | D | |
| 8. | A | |
| 9. | A | |
| 10. | C | |
| 11. | C | |
| 12. | D | |
| 13. | D | |
| 14. | C | |
| 15. | A | |
| 16. | A | |

- | | | |
|-----|--|---|
| 17. | (a) the radiation emitted by a perfect emitter or a perfect absorber | 1 |
| | (b) wavelength / λ ; | 1 |
| | (c) | |



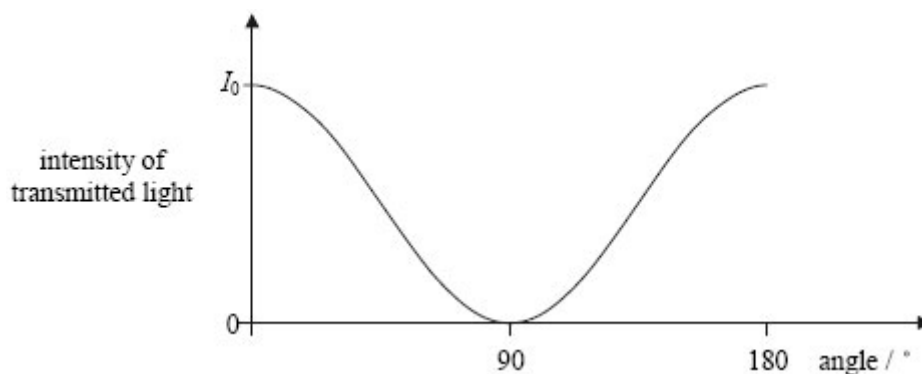
lower intensities with peak shifted to the longer wavelength;

- | | | |
|-----|---|---|
| (d) | $T = \frac{2.90 \times 10^{-3}}{\lambda} = \frac{2.90 \times 10^{-3}}{9.70 \times 10^{-7}} = 3000 \text{ K};$ | 1 |
|-----|---|---|

[4]

- | | | |
|-----|---|---|
| 18. | (a) light where the direction of the (electric) field is always (or predominantly) in the same plane; | 1 |
| | (b) (i) $I = (I_0 \cos^2 60^\circ) = \frac{I_0}{4};$ | 1 |

(ii)



general $\cos^2$ shape;

1 (WS)

max at $\theta = 0$ and curve touches horizontal axis at $\theta = 90^\circ$;

1 (WS)

- (c) light is (partially) horizontally polarised by reflection;
sunglasses have a transmission axis at 90° to the plane of reflected light
causing the intensity of reflected light to be reduced;

1

1

[6]

19. (a) using $d \sin \theta = m\lambda$ and the idea that max m is achieved when $\sin \theta = 1$;

1

making use of $d = \frac{1}{6.0 \times 10^5}$;

1

to give $m = 2.8$;

1

hence there are five complete orders; (2 orders on each side including zeroth)

1

- (b) Spectrum/lines are brighter in the first order;
Lines more likely to be seen separately in second order;

1

1

[6]

20. (a) the direction of the planet's trajectory is changing hence the velocity
of the planet is changing;

1

- (b) (i) arrow pointing vertically downwards;

1

(ii) weight (or gravity);
normal force/reaction force;

1

1

(iii) loss in PE = $0.05 \times 9.8 \times (0.8 - 0.35)$;
= gain in KE = $\frac{1}{2}mv^2$;

1

to give $v = 2.97 \text{ m s}^{-1}$ (3.0 m s^{-1});

1

or

use of $v = \sqrt{2gh}$ to give $v = 4.0 \text{ m s}^{-1}$ at point B;

and then use of $v^2 - u^2 = 2gh$ with $u = 4.0 \text{ m s}^{-1}$ and $h = 0.35 \text{ m}$;
 $g = -9.8 \text{ m s}^{-2}$ to get $v = 3.0 \text{ m s}^{-1}$;

(iv) recognise that resultant force = $\frac{mv^2}{r} = \frac{(0.05 \times 9.0)}{0.175} = 2.6 \text{ N}$;

1

$N = \frac{mv^2}{r} - mg$; (must be stated)

1

$= 2.6 - 0.5 = 2.1 \text{ N}$;

1

[9]

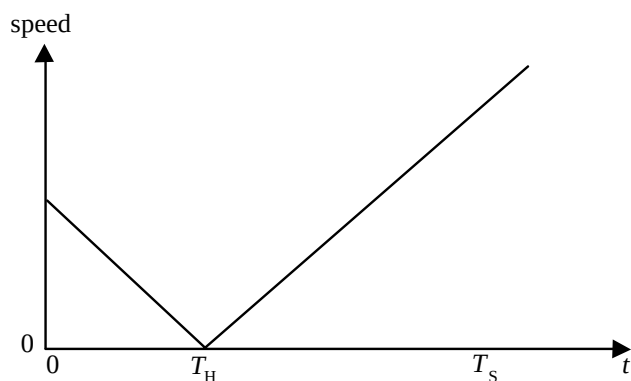
21. (a) initial KE of stone + loss in PE = gain in KE (must indicate this relationship);

1

$\frac{14^2}{2} + 9.8 \times 28 = \frac{v^2}{2}$ to give $v = 27 \text{ m s}^{-1}$ (mass cancels out);

1

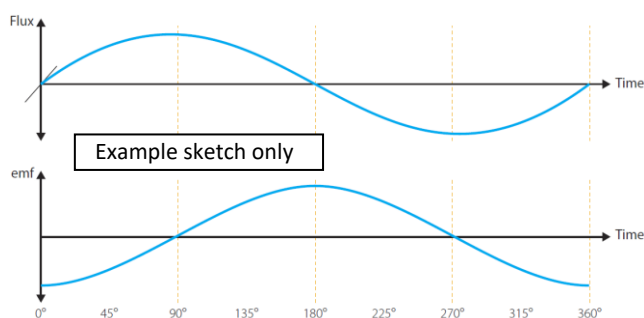
(b)



two lines going in the correct direction as shown; 1 (WS)
 speed greater at T_S than initial speed and zero at T_H (slope magnitudes equal); 1 (WS)

[4]

22. (a) (i) emf (induced) is proportional to; 1 (WS)
 rate of change / cutting of (magnetic) flux; 1 (WS)
- (ii) magnetic field / flux through coil will change as the current changes; 1 (WS)
- (b) (i) sinusoidal and in phase with current; 1 (WS)
 (ii) sinusoidal and same frequency; 1 (WS)
 with 90° phase difference to graph for ϕ and Lenz's law implemented; 1 (WS)



- (iii) emf is reduced; 1 (WS)
 because B is smaller; 1 (WS)
- (c) advantage: no direct contact with cable required; 1 (WS)
 disadvantage: distance to wire must be fixed; 1 (WS)

[10]

23. Kepler's third law

- (a) (i) the orbits are elliptical / not circular; 1
Do not accept "because R changes".
- (ii) gravity / gravitational; 1
- (iii) $F = G \frac{M_s m}{R^2}$; 1

Same symbols as in question must be used to receive the mark.

- (iv) let v = the speed of planet, the acceleration is then = $\frac{v^2}{R}$ 1

$$\text{from Newton 2, } F = G \frac{M_s m}{R^2} = \frac{mv^2}{R}; \left\{ \begin{array}{l} \text{Some reference to Newton 2} \\ \text{is required to receive this mark.} \end{array} \right. \quad 1$$

$$v = \frac{2\pi R}{T}; \text{ therefore, } = G \frac{M_s m}{R^2} = \frac{4\pi^2 m R^2}{T^2}; \quad 1$$

$$\text{therefore, } T^2 = \frac{4\pi^2}{GM_s} R^3; \text{ so } K = \frac{4\pi^2}{GM_s} \quad 1$$

(b) (i) recognise that gravitational field strength = $\frac{v^2}{R} = \frac{4\pi^2 R}{T^2}; \quad 1$

$$= \frac{40 \times 1.1 \times 10^9}{(6.2 \times 10^5)^2}; = 0.1 \text{ N kg}^{-1} \quad 1$$

(ii) $T^2 = \frac{4\pi^2}{GM_s} R^3$ therefore, $M_{jup} = \frac{4\pi^2 R^3}{GT^2}; \quad 2$

$$M_{jup} = \frac{4\pi^2 \times (1.1)^3 \times 10^{27}}{6.7 \times 10^{-11} \times (6.2 \times 10^5)^2}; \approx 2.0 \times 10^{27} \text{ kg}; \quad 1$$

Or

$$0.1 = \frac{GM}{R^2}; M = \frac{0.1R^2}{G}; = \frac{0.1 \times (1.1 \times 10^9)^2}{6.7 \times 10^{-11}} \approx 2.0 \times 10^{27} \text{ kg};$$

[12]

24.

Historical method to measure speed of light experimentally	Marks
Provides a basic, labelled sketch of the procedure used	1(WS)
Describes the basic operation of the method. (Determination of speed purely from theory such as from Maxwell's laws is not accepted)	1(WS)
Explains the physics principles behind the method (example: reflection of light, law of reflection, finite speed of light, orbital motion of planets/moons, eclipses, light aberration, interference patterns, angular rotation/speed, etc.) Need to list two physics principles at least.	2(WS)
Describes two limitations of the technique/results obtained (example: slow human reaction time, short distances for light travel, limitation of timing devices, irregular orbits, etc)	2(WS)

Relationship of speed of light to time and distance	Marks
Explains that the metre is defined as the length of the path travelled by light in a vacuum in 1/c seconds	1(WS)
Explains that one second is defined as the time taken for a fixed number (9,192,631,770) of energy transitions in a Caesium 133 atom. (Note: exact number of transitions are not required but Caesium 133 atom must be stated)	1(WS)

[8]